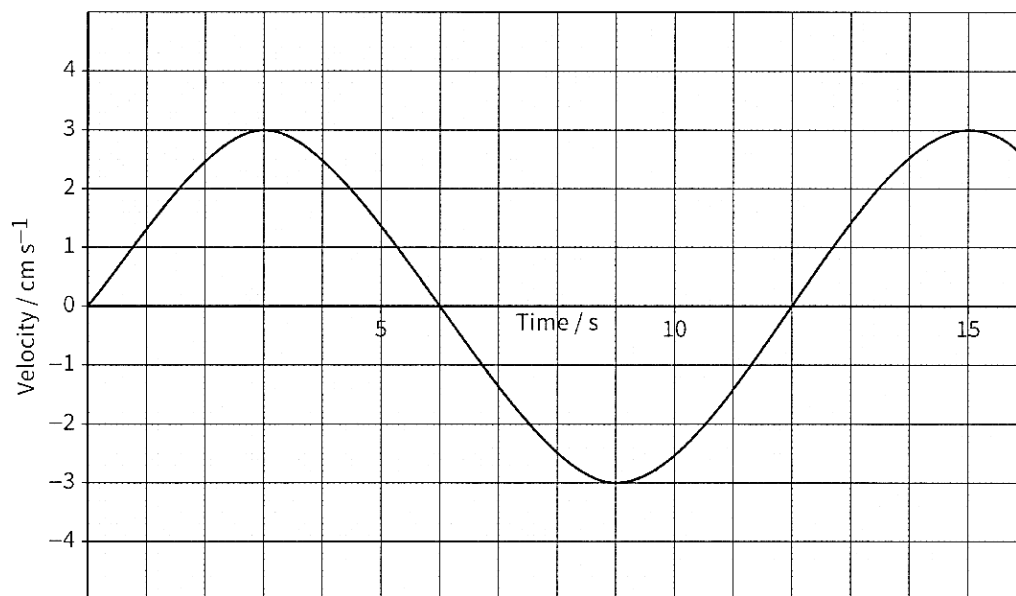


12. The graph below shows the variation with time of the velocity of a 5.0 kg oscillating mass.



What is the maximum restoring force acting on the mass as it oscillates?

- A** 0.041 N **B** 0.050 N **C** 0.079 N **D** 0.15 N

Ans: C

From the graph, period $T = 12$ s, so $\omega = \frac{2\pi}{T} = \frac{2\pi}{12}$.

Also, $v_0 = 3$ cm s⁻¹

Using $v_0 = \omega x_0$, $3 = \frac{2\pi}{12} x_0$ giving $x_0 = 5.730$ cm

Using $|a_0| = \omega^2 x_0$, $a_0 = \left(\frac{2\pi}{12}\right)^2 \times 5.730 = 1.571$ cm s⁻² = 1.571×10^{-2} m s⁻²

Hence maximum restoring force = $ma_0 = 5.0 \times 1.571 \times 10^{-2} = 7.9 \times 10^{-2}$ N

13 An object of mass m oscillates in simple harmonic motion between two springs as